



narrow-band HPM weapons, these have shorter operational ranges.

Components of an HPM weapon system

As outlined earlier, an HPM weapon system essentially comprises a pulsed power source, a source of microwave energy and a transmitting antenna (Fig. 1). The pulsed power source further comprises a power supply and a pulse generator.

Pulse power generators. Pulse power generators that drive HPM sources are generally required to deliver short, intense electrical pulses of 1MV or more with pulse duration up to 1µs. One way to generate the

theoretically be equal to $n \times V$.

Spark gaps are used as switches and breakdown voltage of spark gaps is kept higher than the voltage V across each capacitor. Initially, all capacitors are in parallel and are charged to a voltage V . Spark gaps are in open state. In order to initiate discharge, the first spark gap is externally triggered to the breakdown state connecting the first two capacitors in series and thereby raising the voltage across the second spark gap to $2V$.

The second gap also goes to breakdown state and the process continues till a voltage pulse with amplitude equal to $n \times V$ is applied across the load.

One such Marx generator designed to drive HPM sources is APELC MG20-22C-2000PF from Applied Physical Electronics LC. This megavolt-class Marx generator with its 18-ohm source impedance is specifically designed to drive low-impedance loads. With a 50kV charge voltage it delivers a 500kV, 1.1kJ

pulse into a matched load with a peak power of more than 12GW. This generator uses low-impedance, parallel-switched topology, which makes it well-suited for a wide variety of HPM applications.

Another Marx generator from the same company is MG30-3C-100nF (Fig. 3) that is capable of storing a maximum of 1.8kJ and can deliver 300kV to a matched load. This Marx generator has low impedance of 33ohms and is axially compact in order to drive HPM antennae on remote platforms. Maximum peak power and repetition rate specifications are 5GW and 10Hz, respectively.

The other method is to use a flux-compression generator (Fig. 4). In such a generator, a magnetic coil is compressed either by explosive or

magnetic forces leading to rapidly-rising current pulse. In an explosively-driven flux-compression generator, chemical energy of the explosives is to be partially converted into the energy of an intense magnetic field surrounded by a correspondingly large electric current.

Explosively-driven flux-compression generators are for one-time use only as the equipment gets destroyed during operation.

Such a flux-compression generator constitutes an important component of an e-bomb. It consists of a metal cylinder called armature, which is surrounded by a coil of wire called stator winding. The armature cylinder is filled with high explosive and a sturdy jacket surrounds the entire device. The stator winding and the armature cylinder are separated by empty space. A Marx bank type power source used to drive the stator winding is also a part of the bomb.

An electrical current through the stator winding generates an intense magnetic field. The explosive, when triggered, travels as a wave through the middle of the armature cylinder, forcing it to come in contact with the stator winding, thereby creating a short-circuit. The moving short-circuit compresses the magnetic field, thereby generating an intense electromagnetic burst.

Another common option is to replace the flux-compression generator by a pulse-forming network.

Caution may be exercised to ensure that the pulsed power source delivers a well-matched signal with regards to voltage and impedance. An impedance mismatch between the driver signal and the HPM source leads to poor energy transfer to the HPM source, more so if impedance of the source was lower than impedance of the driver.

Choice of the pulsed power source is therefore largely governed by the type of HPM source to be used. Conventional microwave sources used in radars, etc and HPM

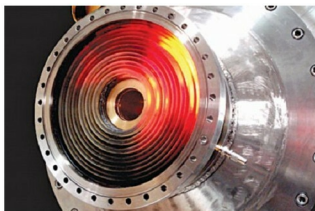


Fig. 3: Marx generator MG30-3C-100nF

required pulses is by using capacitor banks that transform a slowly-rising low-voltage signal into a fast-rising high-voltage signal. A common capacitor bank configuration is Marx bank (Fig. 2) where the capacitors in the bank are connected in parallel during the charging process and switched to a series connection during discharge.

The series connection multiplies the voltage by the number of capacitors in Marx bank. Resistive charging is slow and therefore limits repetition rate. Inductors could also be used in place of resistors. Inductor charging is preferred for higher repetition rates of a few hertz or more. If Marx bank has n capacitors with each charged to a voltage V from a DC power supply, voltage delivered to the load during discharge would